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Chapter 11

The Efficient Market Hypothesis

INVESTMENTS
THIRTEENTH EDITION
BODIE, KANE, MARCUS

Overview

- Maurice Kendall (1953) found no predictable pattern in stock price changes.
    - Prices were as likely to go up as to go down on any particular day, regardless of past performance.
    - How do we explain random stock price changes?

Efficient Markets

- **Efficient market hypothesis (EMH)**
  - Prices of securities **fully reflect available information**.
  - Investors buying securities in an efficient market should expect to obtain an equilibrium rate of return.

Random Walks

- Stock prices should follow a **random walk**
  - Stock price changes are random and unpredictable.
  - Necessary consequence of intelligent investors competing to discover relevant information on which stocks to buy or sell before the rest of the market becomes aware of that information.

Note: Strictly speaking, stock prices follow a submartingale, the expected price change can be positive—compensation for TVM and systematic risk. A random walk is more restrictive—successive stock returns are independent *and* identically distributed.

Figure 11.1 Cumulative Abnormal Returns Before Takeover Attempts

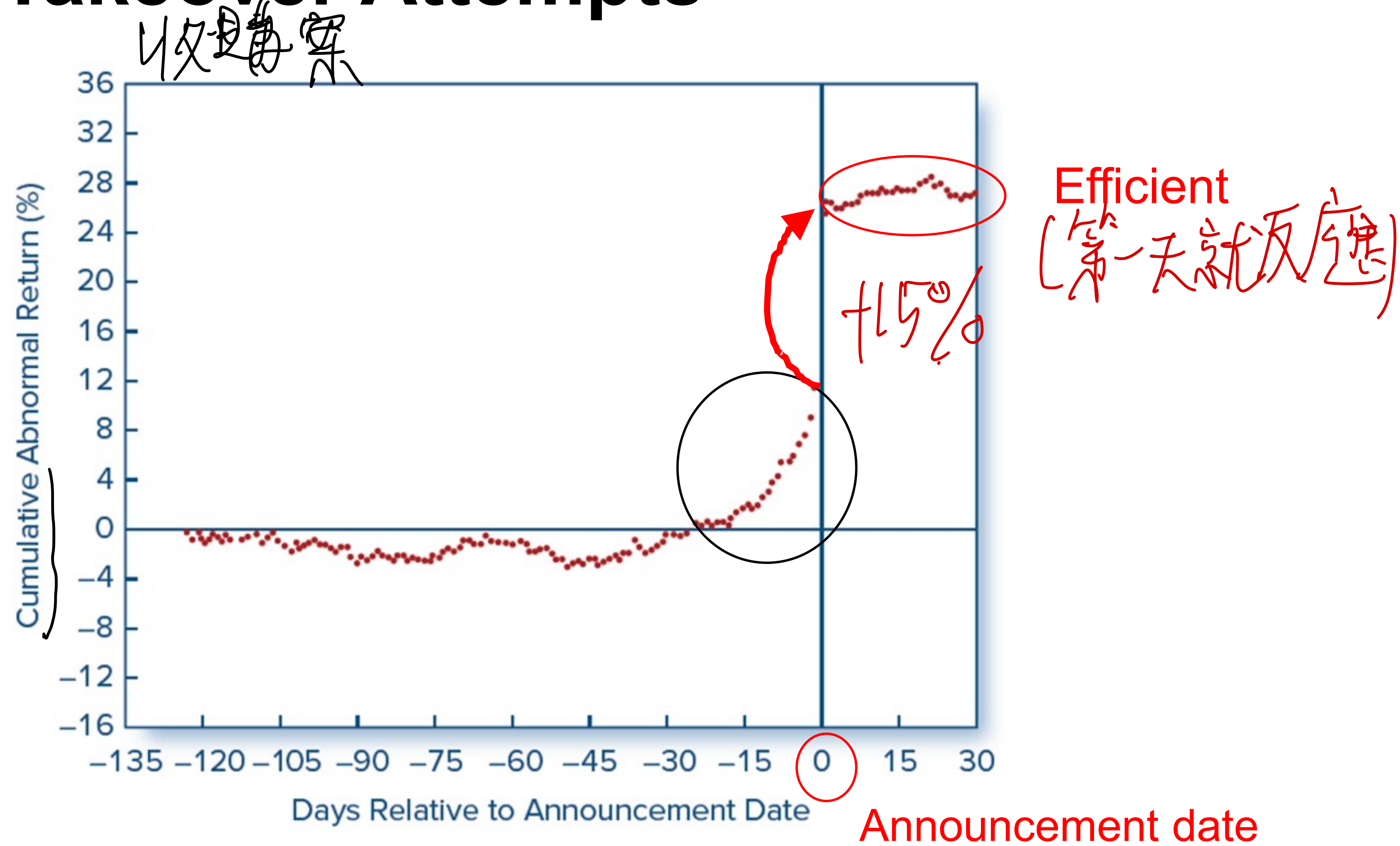


Figure 11.1 Cumulative abnormal returns before takeover attempts: target companies.

Source: This is an update of a figure that appeared in Keown, A., & Pinkerton, J. (1981, September). Merger announcements and insider trading activity. *Journal of Finance, 36*. Updates courtesy of Jinghua Yan.

Stock Price Reaction to CNBC Reports

新聞對價格影響

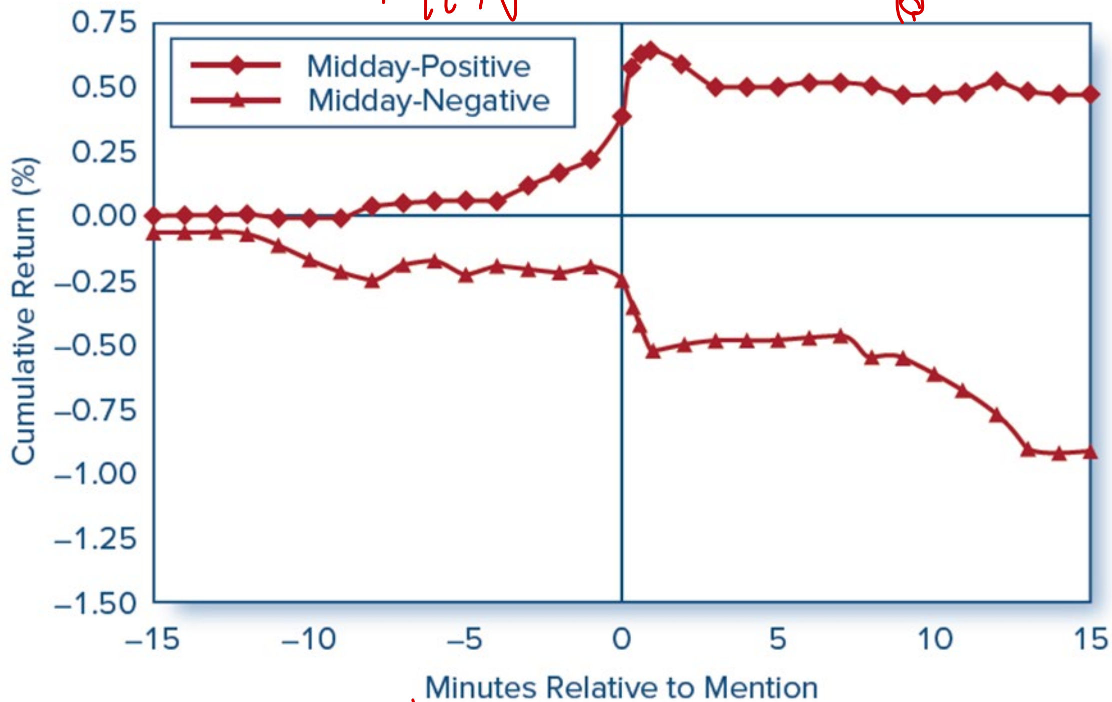


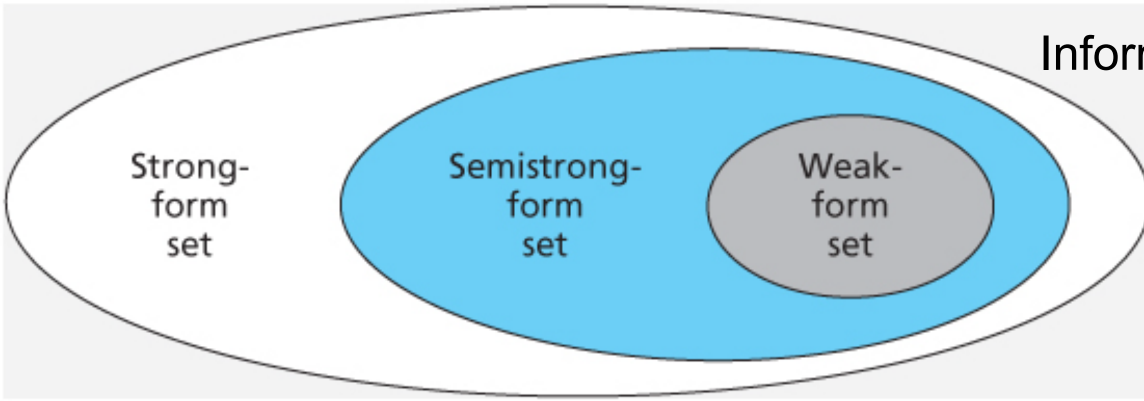
Figure 11.2 Stock price reaction to CNBC reports. The figure shows the reaction of stock prices to on-air stock reports during the "Midday Call" segment on CNBC. The chart plots cumulative returns beginning 15 min before the stock report.

Source: Busse, J. A., & Green, T. C. (2002). Market efficiency in real time. *Journal of Financial Economics, 65*, 422.

Competition as the Source of Efficiency

- Information:
  - The most precious financial commodity.
  - Strong competition assures prices reflect information.
  - Higher investment returns motivates information-gathering.
  - Diminutive marginal returns on research activity suggest only managers of the largest portfolios will find it useful pursuing.

Versions of the EMH



Information of three forms of EMH

All versions assert that prices should reflect available information.

1. **Weak-form** asserts that stock prices already reflect all information contained in the history of past prices.
2. **Semi strong-form** asserts that stock prices already reflect all *publicly* available information.
3. **Strong-form** asserts that stock prices reflect all relevant information, including insider information.

Technical Analysis

- **Technical analysis**—Research to identify mispriced securities that focuses on recurrent and predictable stock price patterns and on proxies for buy or sell pressure in the market.
  - Key to success is a *sluggish response* of stock prices to fundamental supply-and-demand factors.
    - Relative strength
    - Resistance levels, support levels
  - Weak-form EMH implies technical analysis should be fruitless.

Fundamental Analysis

- **Fundamental analysis**
  - Assessment of firm value that focuses on such determinants as earnings and dividends prospects, expectations for future interest rates, and risk evaluation
  - Seeks to find firms that are *mispriced*
    - Attempt to find firms that are better than everyone else's estimate or troubled firms that may be great bargains
- Semi strong-form EMH predicts that *most* fundamental analysis should be fruitless.

Active Versus Passive Portfolio Management

Active Management

- Attempts to beat the market by timing the market or through superior security selection.
- An expensive strategy.
- Suitable for very large portfolios.

Passive Management.

- No attempt to outsmart the market.
- Accept EMH.
- Index Funds and ETF’s.
- Low-cost strategy.

Portfolio Management in an Efficient Market

- Even if the market is efficient, a role exists for portfolio management:
  - Diversification.
  - Tax considerations.
  - Risk profile of investor.
    - For example, investors of different ages.
  - In an EM: Tailor the portfolio to investors (according to age, tax bracket, risk aversion, and employment), rather than beat the market.

Resource Allocation

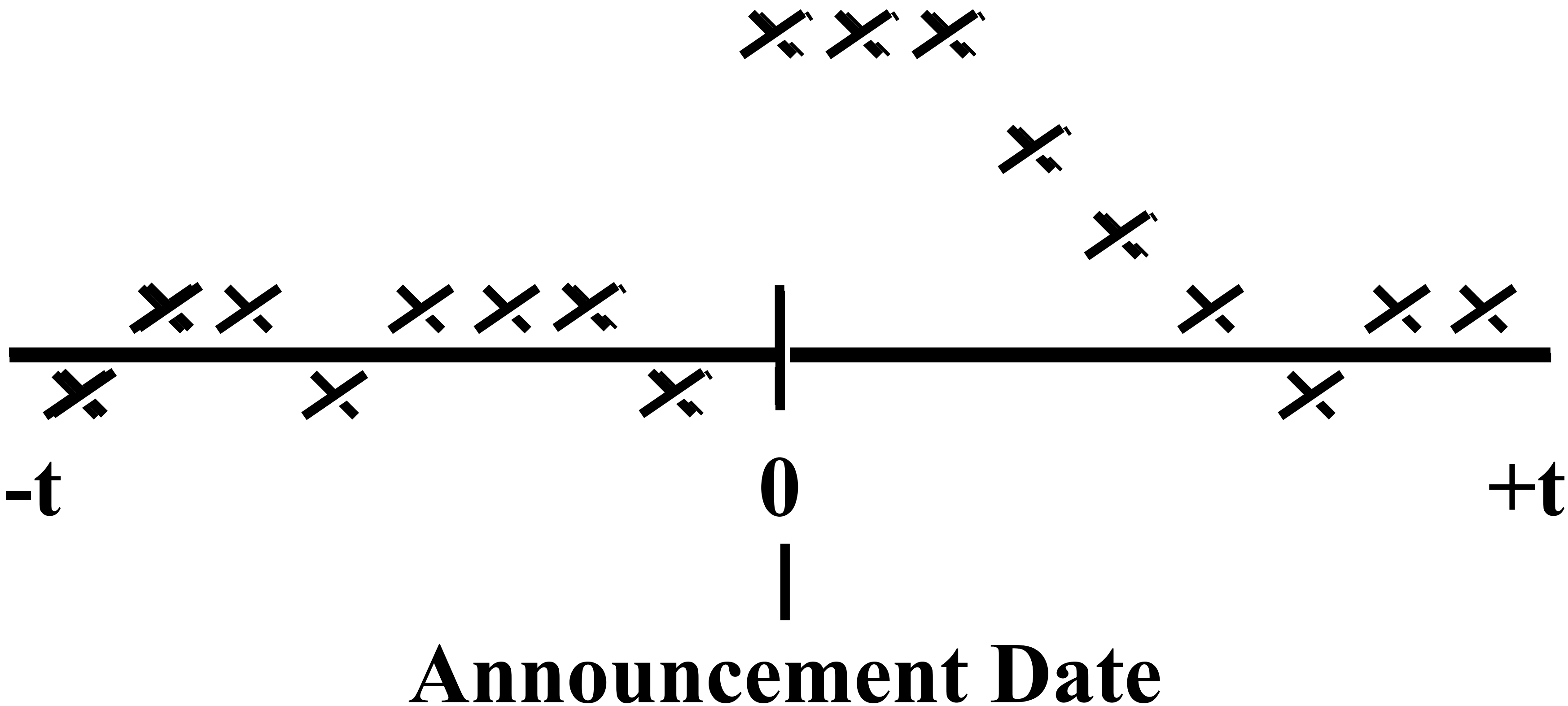
- Inefficient markets result in systematic resource misallocation.
  - Overvalued securities can raise capital too cheaply.
  - Corporations with undervalued securities may pass up profitable opportunities because the cost of raising capital is too high.
  - Efficient market ≠ perfect foresight market.

Event Studies 事件分析

- An **event study** is a methodological approach designed to measure the impact of an event of interest on stock returns.
- The **abnormal return** due to the event is the difference between the stock's actual return and a ***proxy*** for the stock's return in the absence of the event.

How Tests Are Structured
(1 of 2)

1. Examine prices and returns over time



How Tests Are Structured
(2 of 2)

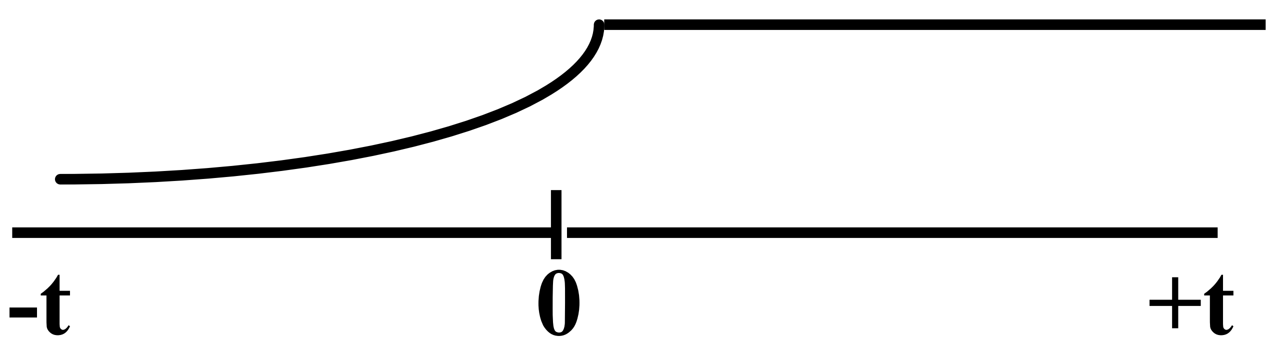
2. Returns are adjusted to determine if they are abnormal

- Market Model approach:

a. Expected Return: $r_t = \alpha + \beta r_{Mt} + e_t$

b. Abnormal Return: $e_t = r_t - (\alpha + \beta r_{Mt})$

c. Cumulate the abnormal returns over time:
(CAR: information leakage)



Example 11.4 Abnormal Returns

- Suppose that the analyst has estimated that a α=0.05% and β=0.8.
- On a day that the market goes up by 1%, you would predict from market model that the stock should rise by and expected value of

$$0.05\% + 0.8(1\%) = 0.85\%$$

- If the stock actually rises by 2%, the analyst would infer that firm-specific news that day caused an additional stock return of

$$2\% - 0.85\% = 1.15\% \text{ (abnormal return for the day)}$$

Example 11.5 Using Abnormal Return to Infer Damage

- Suppose the stock of a company with market value of \$100 million falls by 4% on the day that news of an accounting scandal surfaces.
- The rest of the market generally did well that day. The market indexes were up sharply.
- Based on the usual relationship between stock and the market, one would have expected a 2% gain on the stock.
- We would conclude that the impact of the scandal was a 6% drop in value.
- One might then infer that the damages sustained from the scandal were \$6 million (= \$100 million x 6%) when investors reassessed the value of the stock.

Are Markets Efficient?

- **Magnitude Issue.**
  - Only managers of large portfolios earn enough trading profits to make exploiting minor mispricing worth the effort.
- **Selection Bias Issue.**
  - Only unsuccessful (or partially successful) investment schemes are made public; good schemes remain private.
- **Lucky Event Issue**
  - For every big winner, there may be many big losers, but we never hear of these managers.

Weak-Form Tests

- Returns over *short* horizons.
  - Serial Correlation: Small positive serial correlation of weekly returns of NYSE stocks
  - Tendency of poorly performing stocks and well-performing stocks in one period to continue that abnormal performance in following periods is the **momentum effect**. (using 3- to12-month holding period; Jegadeesh and Titman)

Weak-Form Tests

- Returns over *long* horizons
  - **Reversal effect** is the tendency of poorly performing stocks and well-performing stocks in one period to experience reversals in following periods. (DeBondt and Thaler (1985))
  - Rank the performance of stocks over a 5-year period (25% cumulative return in the following 3-year period). Imply contrarian investment strategy.
  - Episodes of overshooting followed by correction
  - Shiller points out that overreaction followed by correction would cause prices to fluctuate around fair value (Figure 11.3).

Figure 11.3 Prices Fluctuate Around Intrinsic Value

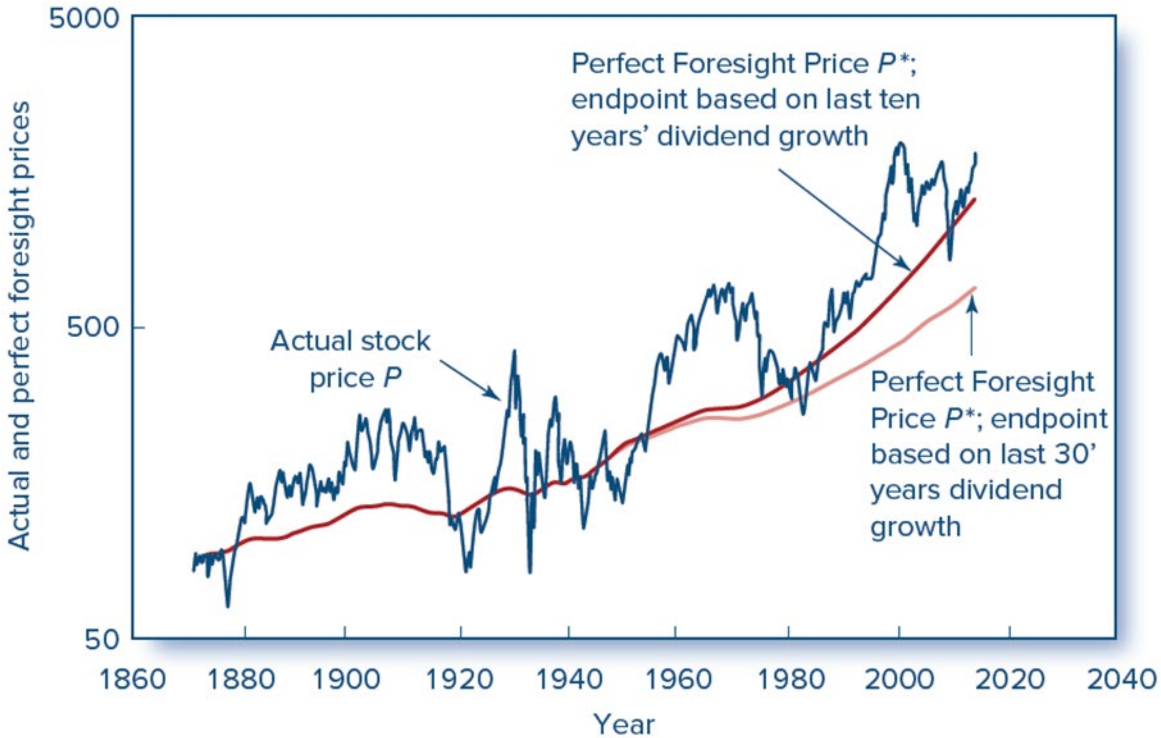


Figure 11.3 Real (inflation-adjusted) value of S&P500 and two estimates of intrinsic value obtained by discounting future dividends plus terminal value in 2013 based on a constant-growth dividend discount model. Discount rate equals 7.6%, the historical average real market return since 1871.

Source: Shiller, R. J. Speculative asset prices, revision of 2013 Nobel Prize Lecture, available at **www.nobelprize.org/uploads/2018/06/shiller-lecture.pdf.**

Predictors of Broad Market Returns

- Fama and French.
  - Return on the aggregate stock market tends to be higher when the dividend yield is high.
- Campbell and Shiller.
  - Earnings yield can predict market returns.
- Keim and Stambaugh.
  - Bond spreads can help predict broad market returns.

Predictors of Broad Market Returns

- These variables are proxying for variation in the market risk premium.
- Given a level of dividends or earnings yield, stock prices will be lower and dividend and earnings yields will be higher when the risk premium (and therefore the expected market return) is higher.
- Predictability in the risk premium, not in risk-adjusted abnormal returns (alpha)

Semistrong Tests

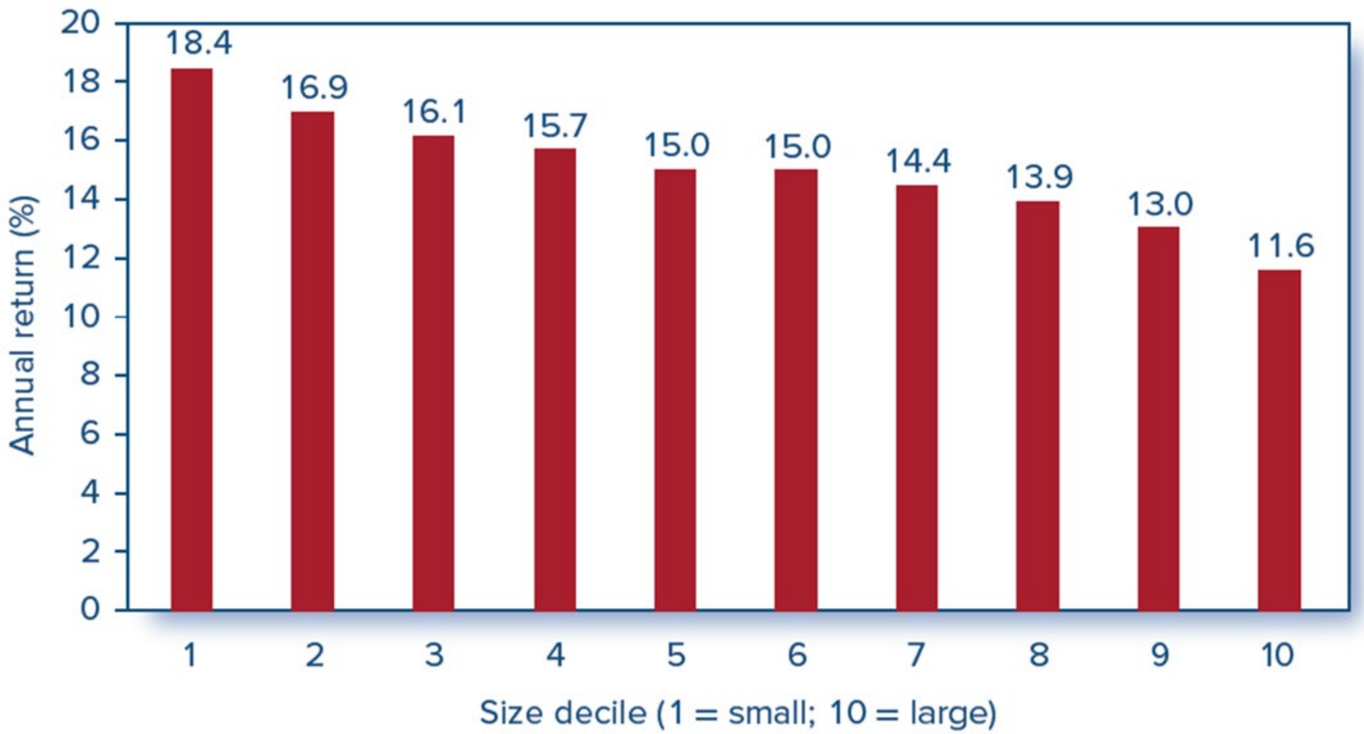
- Efficient market ***Anomalies***: Patterns of returns that seem to contradict the efficient market hypothesis
- Need to adjust for portfolio risk before evaluating the success of an investment strategy
  - High return stocks may just have high market betas
- Tests of risk-adjusted returns are joint tests of the EMH and the risk adjustment procedure

Semi-Strong Tests: Anomalies

- ***P*/*E* effect** – Basu discovered that portfolios of low-*P*/*E* ratio stocks have provided higher returns than high *P*/*E* portfolios.

- **Small-firm effect** – Stocks of small firms can earn abnormal returns, primarily in January

Figure 11.4 Semi-Strong Tests: Small-Firm Effect



Small Firm Effect:

(January effect)

Figure 11.4 Average annual return for 10 size-based portfolios, 1926–2021.

Source: Authors' calculations, using data obtained from Professor Ken French's data library at **http://mba.tuck .dartmouth.edu/pages/faculty/ken.french/data_lIbrary.html.**

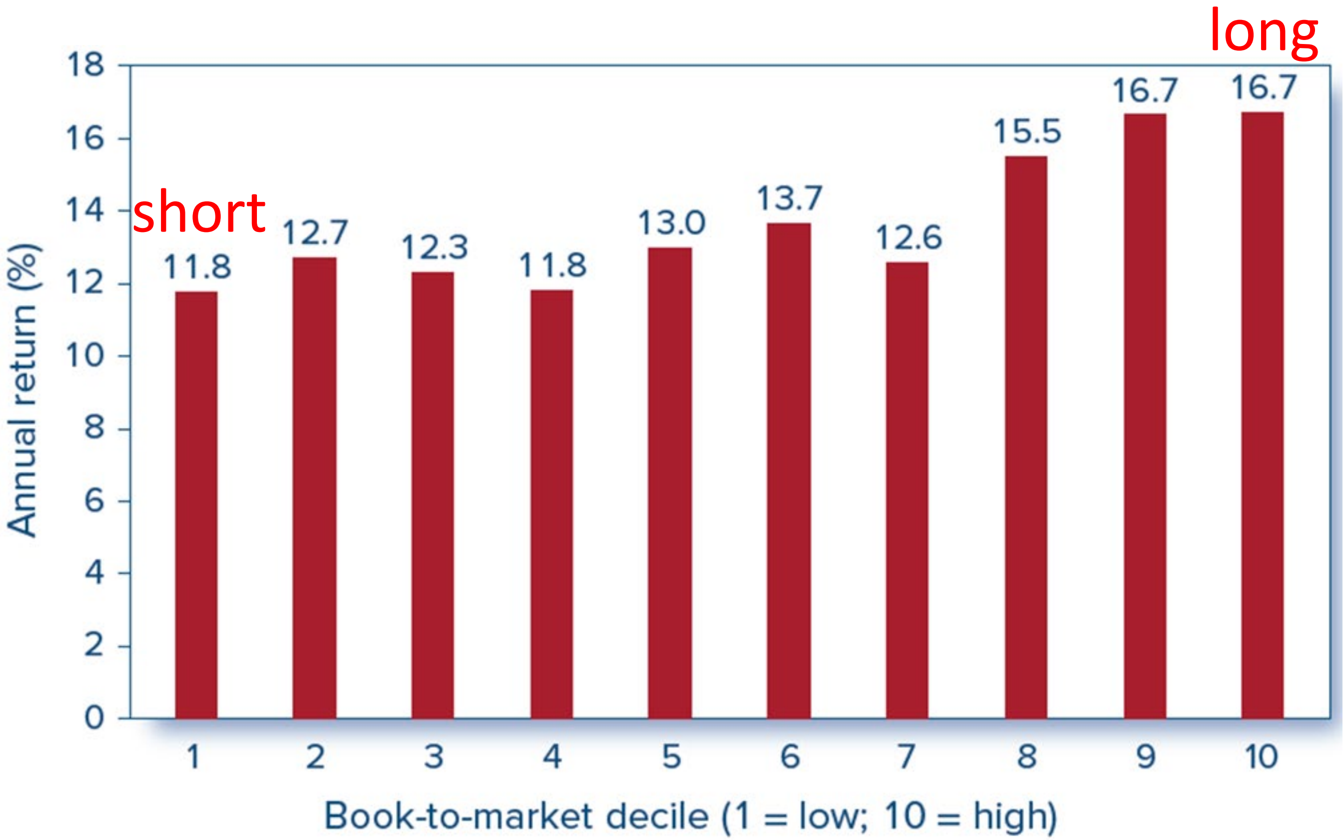
Semi-Strong Tests: Anomalies

- **Neglected-firm effect** – investments in stock of less well-known firms have generated abnormal returns. (# of institutions holding, small and illiquid stock)
  - Liquidity effect – Illiquid stocks have a strong tendency to exhibit abnormally high returns

- **Book-to-market effect** – Shares of high book-to-market firms can generate abnormal returns

Figure 11.5 Semi-Strong Tests: Book to Market Effects

long-short trading strategies



Book-to-Market Ratios

Figure 11.5 Average return as a function of book-to-market ratio, 1926–2021.

Source: Authors' calculations, using data obtained from Professor Ken French's data library at **http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_lIbrary.html.**

Figure 11.6 Semi-Strong Tests: Post-Earnings-Announcement Price Drift

Price Drift.

- Ball and Brown find a sluggish response of stock prices to firms' earnings announcements.
- Market appears to adjust to the earnings information only gradually, resulting in a sustained period of abnormal returns. (momentum)

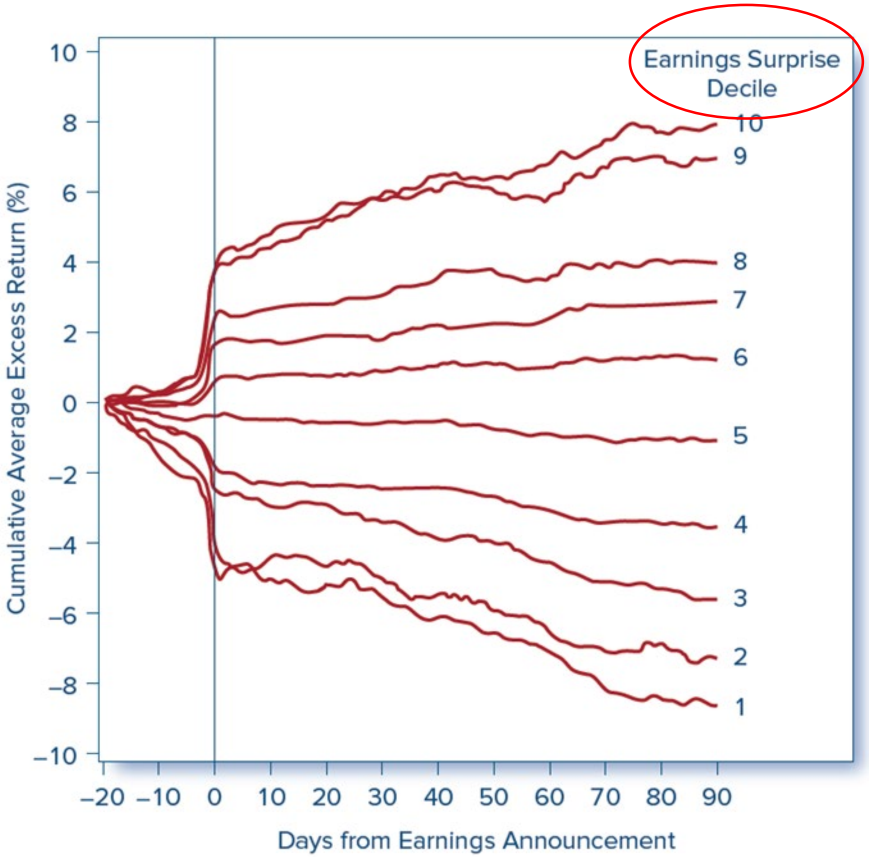


Figure 11.6 Cumulative abnormal returns in response to earnings announcements.

Source: Rendleman, R. J. Jr., Jones, C. P., & H. A. Latané, H. A. (1982). Empirical anomalies based on unexpected earnings and the importance of risk adjustments. *Journal of Financial Economics, 10*, 269–287.

Other Predictors of Stock Returns

- **Volatility:** Idiosyncratic volatility negatively associated with returns.
- **Accruals and Earnings Quality:** High accruals have predicted low future returns.
- **Growth:** More rapidly growing firms (with high capital investments, sales growth, asset growth, or high share issuance) tend to have lower future returns.
- **Profitability:** Gross profitability seems to predict higher stock returns.
- ***q*-factor:** Tobin's *q* is a good predictor of average stock returns. Rational interpretation: High q might imply low discount rate (high present value).

Strong-Form Tests: Inside Information

- The ability of insiders to trade profitability in their own stock has been documented in studies by Jaffe, Seyhun, Givoly, and Palmon.
- SEC requires all insiders to register their trading activity.
  - within two business days
  - Trades become public information
- Do **not** expect markets to be strong-form efficient!

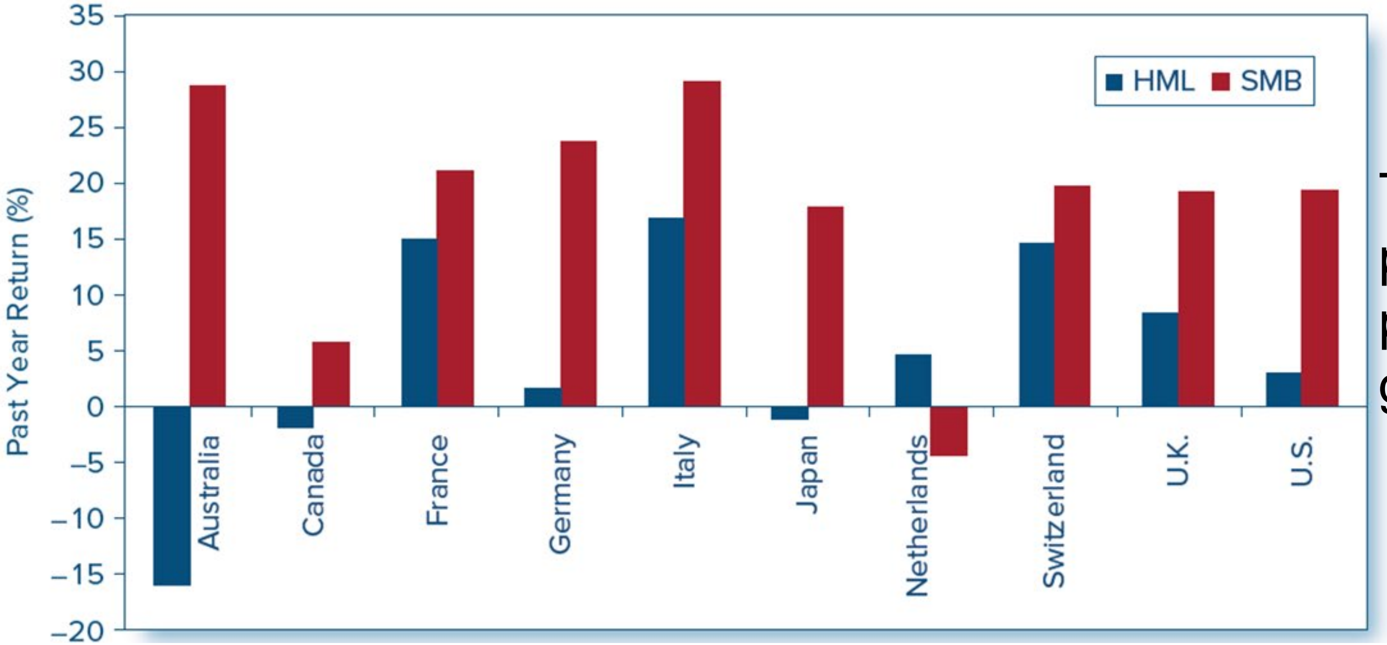
Interpreting the Anomalies

- Feature that small firms, low market-to-book firms, and recent “losers” seem to have in common is a stock price that has fallen considerably in recent months/years.
  - Fama and French argue that these effects can be explained by risk premiums.
  - Lakonishok, Shleifer, and Vishny argue that these effects are evidence of inefficient markets.

Interpreting the Anomalies

- Risk Premiums or market inefficiencies—disagreement here
  - Fama and French argue that these effects can be explained as manifestations of risk stocks with higher betas (size, BM)
  - Proxies for more fundamental determinants of risk
  - The returns on “style” portfolios (SMB, HML) do indeed seem to predict business cycles (Fig. 11.7).

Figure 11.7 Predictors of GDP Growth



Tend to have positive returns prior to rapid growth in GDP

Figure-11.7 Return to style portfolios as predictors of GDP growth. Average difference in the return on the style portfolio in years before good GDP growth versus in years with bad GDP growth. Positive value means the style portfolio does better in years prior to good macroeconomic performance. HML= high-minus-low portfolio, sorted on ratio of book-to-market value. SMB = small-minus-big portfolio, sorted on firm size.

Source: Liew, J., & Vassalou, M. (2000). Can book-to-market, size, and momentum be risk factors that predict economic growth? *Journal of Financial Economics, 57*, 221–245.

Interpreting the Anomalies

- Risk Premiums or market inefficiencies
  - Lakonishok, Shleifer, and Vishney argue that these effects are evidence of inefficient markets
  - Systematic errors in the forecasts of stock analysts. Analysts seem overly pessimistic about firms with low growth prospects and overly optimistic about firms with high growth prospects.
  - When market participants recognize their errors, prices reverses.

Interpreting the Evidence 1

- Anomalies or data mining?
  - Some anomalies have disappeared
  - Book-to-market, size, and momentum may be real anomalies.
- Anomalies over time.
  - Exploiting them eliminates abnormal profits
  - Should self-destruct in well-functioning markets.
  - Chordia, Subrahmanyam, and Tong find evidence that attenuating abnormal returns in the post-1993 period – liquidity and low trading costs facilitate efficient price discovery

Interpreting the Evidence 2

- Bubbles and market efficiency.
- Bubbles occur when a rapid run-up in prices creates a widespread expectation that they will continue to rise.
  - For example, tulip mania in 17th-century Holland.
- Rush to “get in on the action”.
- Inevitably, the run-up stalls and the bubble ends in a crash.
- Bubbles are difficult to predict and exploit

Stock Market Analysts

- Some analysts may add value, but:
  - Tend to be overwhelmingly positive in the assessment of the prospects of firms.
    - On a scale of 1 (strong buy) to 5 (strong sell), the average recommendation for 5,628 covered firms in 1996 was 2.04
  - Difficult to separate effects of new information from changes in investor demand.
  - Recommendations may lead to investing strategies that are too expensive to exploit.

Mutual Fund Performance 1

- Casual evidence does not support the claim that professionally managed portfolios can consistently beat the market.
- Conventional performance benchmark today is a four-factor model.
  - Three Fama-French factors (Market, Size, Book-to-Market)
  - Momentum factor.

Estimates of Individual Mutual Fund Alphas

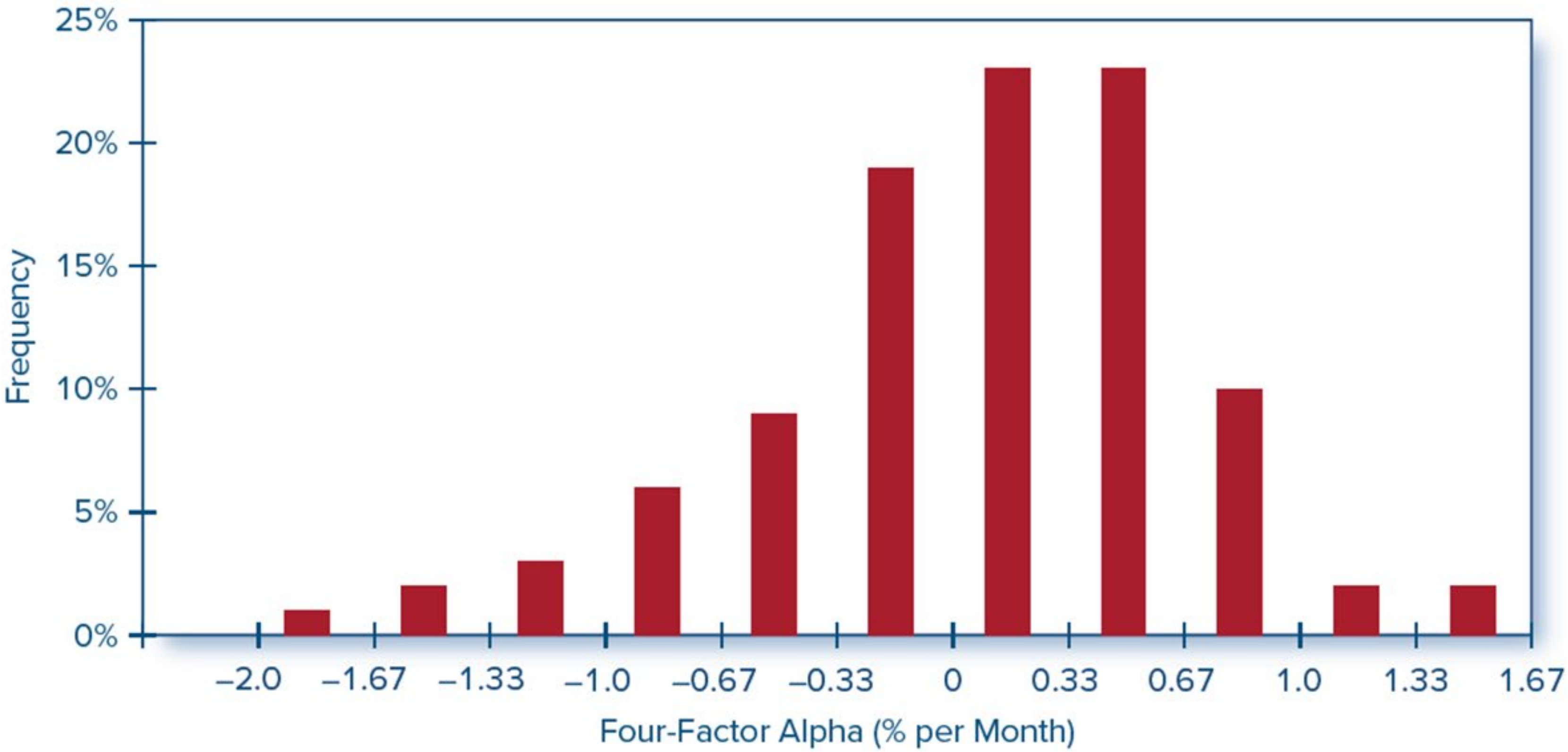


Figure 11.8 Mutual fund alphas computed using a four-factor model of expected return, 1993–2007. (The best and worst 2.5% of observations are excluded from this distribution.)

Source: Professor Richard Evans, University of Virginia, Darden School of Business.

Table 11.1 Consistency of Investments Results

Table 11.1

Consistency of investment results. Percentage of top-half funds that remain in the top half in the following 2 years.

	Number of top-half performers in year ending June 2019	Percentage of 2019 outperformers that perform in top half of sample in year ending June 2020	Percentage of 2019 outperformers that perform in top half of sample in both year ending June 2020 and year ending June 2021
All domestic equity funds	1,041	71.8	19.0 <25%
Large-cap equity funds	373	61.1	25.5
Small-cap equity funds	26.1	69.4	16.5

Source: Liu, B., & Sinha, G. (2021, October). U.S. persistence scorecard, mid-year 2021. S&P Dow Jones Indices.

Mutual Fund Performance 2

- Consistency.
  - Carhart – finds minor persistence in relative performance across managers, but much of that persistence seems due to expenses and transaction costs.
  - Bollen and Busse – support for performance persistence over short horizons.
  - Berk and Green – skilled managers will attract new funds until the costs of managing those extra funds drive alphas down to zero.

Figure 11.9 Risk-Adjusted Performance in Ranking Quarter and Following Quarter

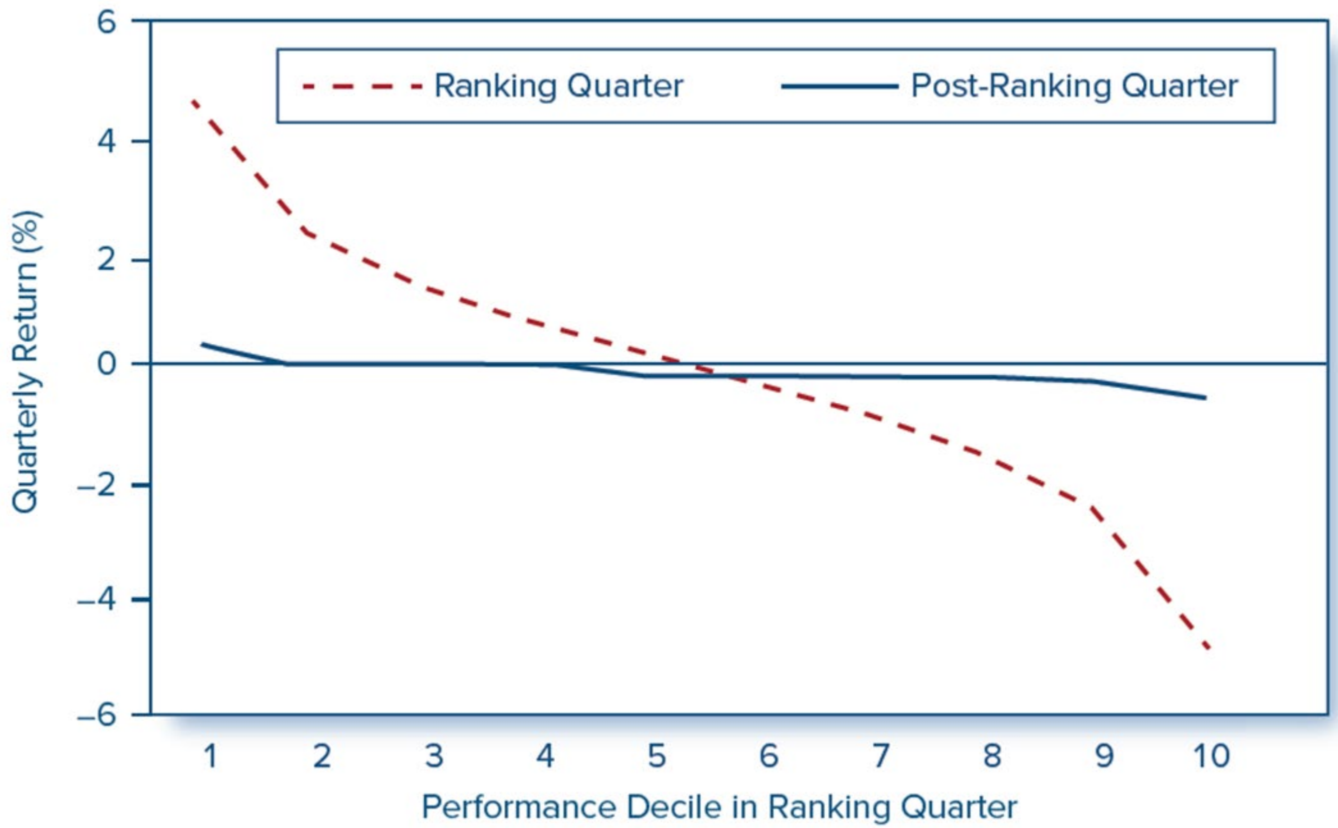


Figure 11.8 Risk-adjusted performance in ranking quarter and following quarter.

So, Are Markets Efficient?

- Enough anomalies exist in the empirical evidence to justify the search for underpriced securities that clearly takes place.
  - However, the market is competitive enough that only differentially superior information or insight will earn money.
  - Margin of superiority that any professional manager can add is so slight that the statistician will not easily be able to detect it.